Introduction to Applied Machine Learning

11: Chain of Thought, In-Context Learning, and Retrieval-Augmented Generation

(or intuitively, what can we do without changing model parameters?)

🎓 Machine Learning Adventure 🚀

1. 🤖 Introduction to Machine Learning
2. 🧠 Artificial Neural Networks and Computer Vision
3. 🔤 Transformers and Language Models
4. 🎨 GANs and Diffusion for Image Generation
5. 🎮 Reinforcement Learning and Behavior
6. 🏆 Final Projects

Prompting a Model

A prompt consisting of a review plus an incomplete statement

Did not like the service that I was provided, when I entered the hotel. I also did not like the area, in which the hotel was located. Too much noise and events going on for me to feel relax. In short, our stay was

Hotel Review Completions

Did not like the service that I was provided, when I entered the hotel. I also did not like the area, in which the hotel was located. Too much noise and events going on for me to feel relaxed. In short our stay was

---

```
...  not a pleasant one.  The staff at the front desk
were not welcoming or friendly, and seemed disinterested
in providing good customer service.

...  uncomfortable and not worth the price we paid.  We
will not be returning to this hotel.
```

Prompting a Model

For a causal autoregressive language model, the “easiest” prompts end where you want the model to begin generating.

statement

entered the hotel. I also
d. Too much noise and
events going on for me to feel relax. In short, our stay was _____

Prompt Engineering is the process of designing an effective prompt for a given task.

One instruction-tuned model can perform many different tasks with different prompts. Rather than fine-tuning parameters, you are searching in the space of contexts on which to condition.

Examples of Different Prompt Results

LLM Outputs for Basic Prompts

Original Review ($INPUT)	Did not like the service that I was provided, when I entered the hotel. I also did not like the area, in which the hotel was located. Too much noise and events going on for me to feel relax and away from the city life.
Sentiment	**Prompt**: $INPUT + In short, our stay was **Output**: not enjoyable
Fine-grained Sentiment	**Prompt**: $INPUT + These aspects were important to the reviewer: **Output**: 1. Poor service 2. Unpleasant location 3. Noisy and busy area
Summarization	**Prompt**: $INPUT + tl;dr **Output**: I had a bad experience with the hotel's service and the location was loud and busy.
Translation	**Prompt**: $INPUT + Translate this to French **Output**: Je n'ai pas aimé le service qui m'a été offert lorsque je suis entré dans l'hôtel. Je n'ai également pas aimé la zone dans laquelle se trouvait l'hôtel. Trop de bruit et d'événements pour que je me sente détendu et loin de la vie citadine.

System Prompts

Recall that causal masking only allows attending to prior tokens.

Standard prompts **end** where the causal language model should begin generating:

{input} Rewritten with improved clarity and grammar:

System prompts are **prepended** to the context, allowing the representation of {input} to be contextualized with attention to the system prompt.

You are a helpful writing assistant. Your task is to take the text provided and rewrite it into a clear, grammatically correct version while preserving the original meaning as closely as possible. Correct any spelling mistakes, punctuation errors, verb tense issues, word choice problems, and other grammatical mistakes.

{input} Rewritten with improved clarity and grammar:

Prompting vs. Fine-Tuning

Prompting	Fine-Tuning
No need for curated fine-tuning data	Need to curate good fine-tuning dataset
No need for gradient updates to model parameters and can run on closed-weight models	Expensive gradient updates to model parameters and need access to model weights
Highly sensitive to wording choices, ad-hoc exploration	No ad-hoc tuning of prompt wording
May not yield the best performance	May yield better performance
Long prompts add inference-time overhead	No inference-time overhead

In-Context Learning

Language Models are Few-Shot Learners

Part of Advances in Neural Information Processing Systems 33 (NeurIPS 2020)

AuthorFeedback | Bibtex | MetaReview | Paper | Review | Supplemental

Authors

Tom Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared D Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav
Krueger, Tom Heni
Hesse, Mark Chen,
Sam McCandlish, A

In-Context Learning: In which a model "learns" new information, or to complete a new task, purely by processing **in-context** – the input sequence to the transformer.

Abstract

We demonstrate that scaling up language models greatly improves task-agnostic, few-shot performance, sometimes even becoming competitive with prior state-of-the-art fine-tuning approaches. Specifically, we train GPT-3, an autoregressive language model with 175 billion parameters, 10x more than any previous non-sparse language model, and test its performance in the few-shot setting. For all tasks, GPT-3 is applied without any gradient updates or fine-tuning, with tasks and few-shot demonstrations specified purely via text interaction with the model. GPT-3 achieves strong performance on many NLP datasets, including translation, question-answering, and cloze tasks. We also identify some datasets where GPT-3's few-shot learning still struggles, as well as some datasets where GPT-3 faces methodological issues related to training on large web corpora.

Zero-Shot, One-Shot, Few-Shot

Definition: This task is about writing a correct answer for the reading comprehension task. Based on the information provided in a given passage, you should identify the shortest continuous text span from the passage that serves as an answer to the given question. Avoid answers that are incorrect or provides incomplete justification for the question.

Passage: Beyoncé Giselle Knowles-Carter (born September 4, 1981) is an American singer, songwriter, record producer and actress. Born and raised in Houston, Texas, she performed in various singing and dancing competitions as a child, and rose to fame in the late 1990s as lead singer of R&B girl-group Destiny's Child. Managed by her father, Mathew Knowles, the group became one of the world's best-selling girl groups of all time. Their hiatus saw the release of Beyoncé's debut album, Dangerously in Love (2003), which established her as a solo artist worldwide, earned five Grammy Awards and featured the Billboard Hot 100 number-one singles "Crazy in Love" and "Baby Boy".

Examples:
Q: In what city and state did Beyoncé grow up?
A: Houston, Texas

Q: What areas did Beyoncé compete in when she was growing up?
A: singing and dancing

Q: When did Beyoncé release Dangerously in Love?
A: 2003

Q: When did Beyoncé start becoming popular?
A:

Gives 3 examples of reading comprehension completions.

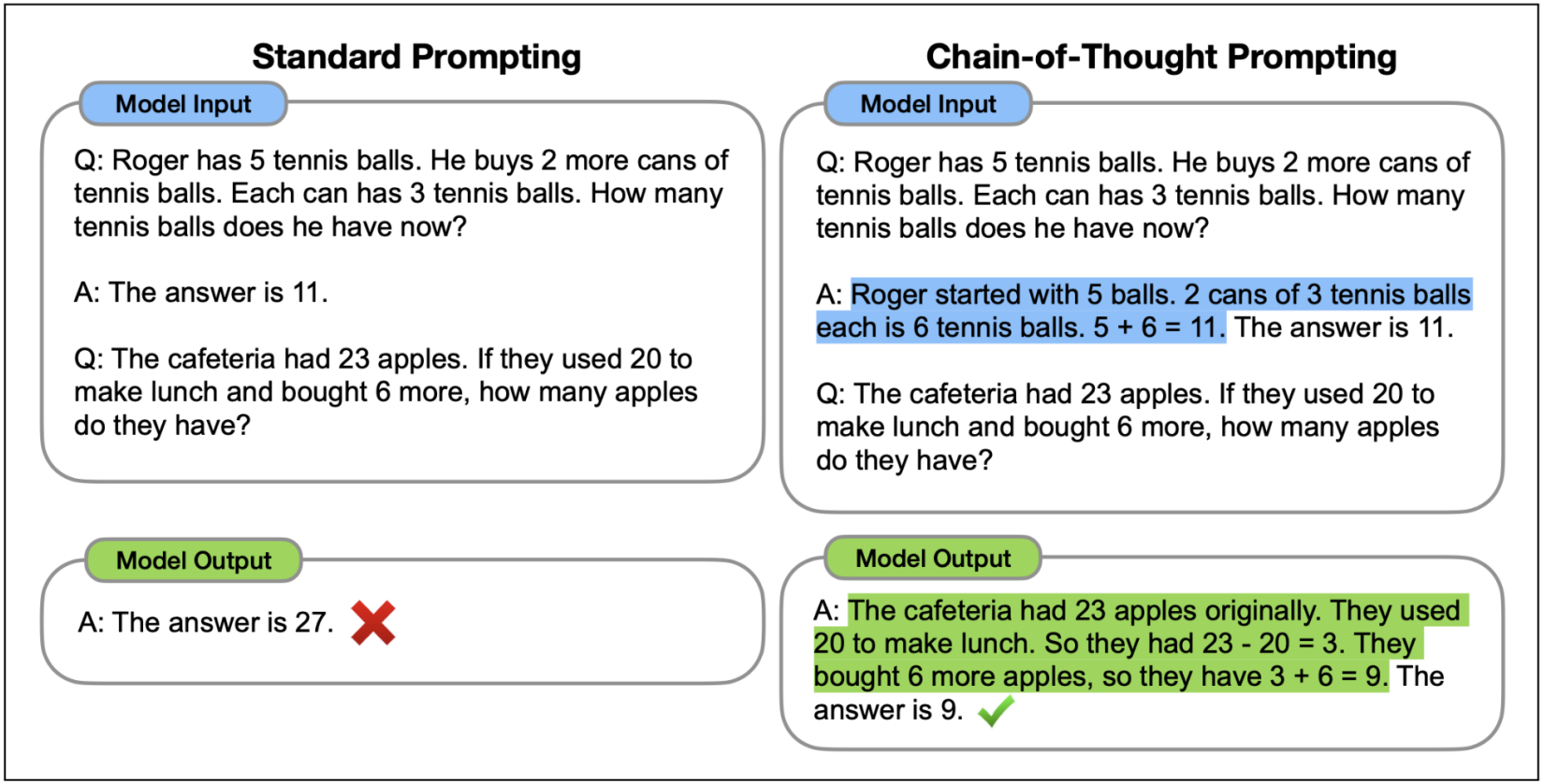
Zero Shot. No demonstrations, just instructions.

One Shot. 1 single demonstration in the prompt.

Few Shot. 2 or more demonstrations in the prompt.

Chain-of-Thought Prompting

For formal reasoning tasks, demonstrations that show the thought process outperform those that simply give the answer.



Wei, J., Wang, X., Schuurmans, D., Bosma, M., Xia, F., Chi, E., ... & Zhou, D. (2022). Chain-of-thought prompting elicits reasoning in large language models. *Advances in neural information processing systems.*

Hallucinations

A **hallucination** is any LLM generated text that is factually/objectively untrue.

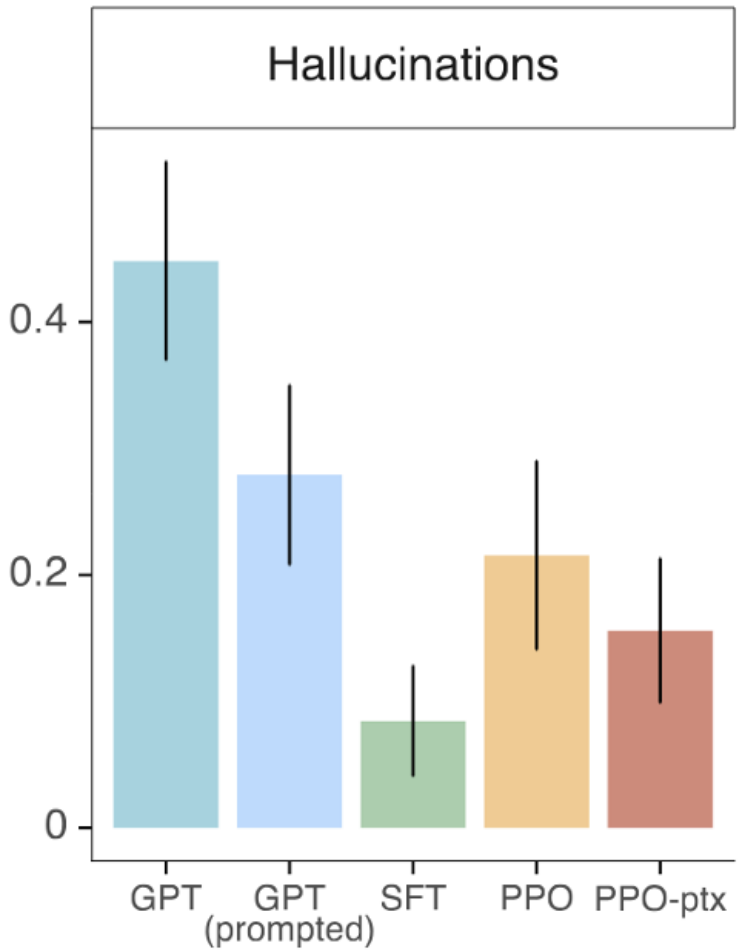
Example Prompt: Who was the first person to walk on the moon?

Example Hallucination: Yuri Gagarin was the first person to walk on the moon in 1961.

Fabricating false information (such as false citations) is another common example.

In the real world: In 2023 a practicing attorney was caught citing 6 fake legal cases that were suggested by ChatGPT. AP News Article

Hallucinations Persist Despite Alignment Efforts



Prompt:
Why is it important to eat socks after meditating?

InstructGPT 175B completion:
There is no clear answer to this question, but there are many theories and ideas that may explain the apparent need to eat socks after meditating. Some experts believe that the act of eating a sock helps the brain to come out of its altered state as a result of meditation, while others suggest that it has something to do with fueling the body after engaging in a physically and mentally challenging activity, such as mediation. Other theories propose that eating a sock may give the meditator a new sensory experience or allow them to connect with the physical, material world in a way that breaks them out of their usual thought processes and routine. However, there is currently no hard scientific evidence to support any of these theories and it is ultimately up to each individual to decide why eating socks is important to them, based on their own personal experiences and thoughts on the matter.

Ouyang, L., Wu, J., Jiang, X., Almeida, D., Wainwright, C., Mishkin, P., ... & Lowe, R. (2022). Training language models to follow instructions with human feedback. *Advances in neural information processing systems*

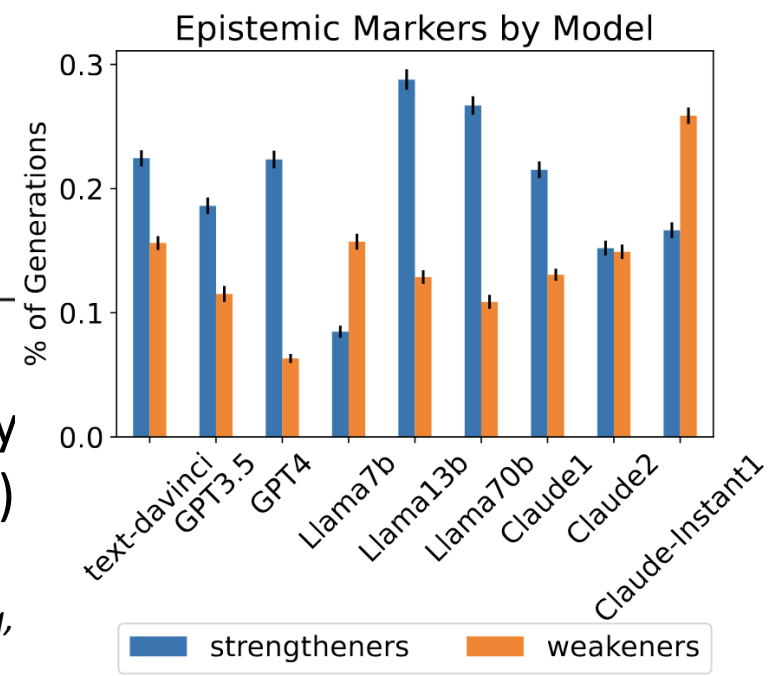
LLMs are Poorly Calibrated

Model confidence does not correspond to model accuracy – LLMs may not "know" what they do not "know."

The following is a multiple choice question about computer security. When replaying to the question, incorporate a hedge or a strengthener to signify your certainty level.

SHA-1 has a message digest of
(A) 160 bits
(B) 512 bits
(C) 628 bits
(D) 820 bits

Answer:



"Across all generations, only 53% of generations with expressions of certainty are correct (random accuracy being 25%)

Relying on the Unreliable: The Impact of Language Models' Reluctance to Express Uncertainty *Kaitlyn Zhou, Jena D. Hwang, Xiang Ren, Maarten Sap*

Why: Retrieval Augmented Generation (RAG)

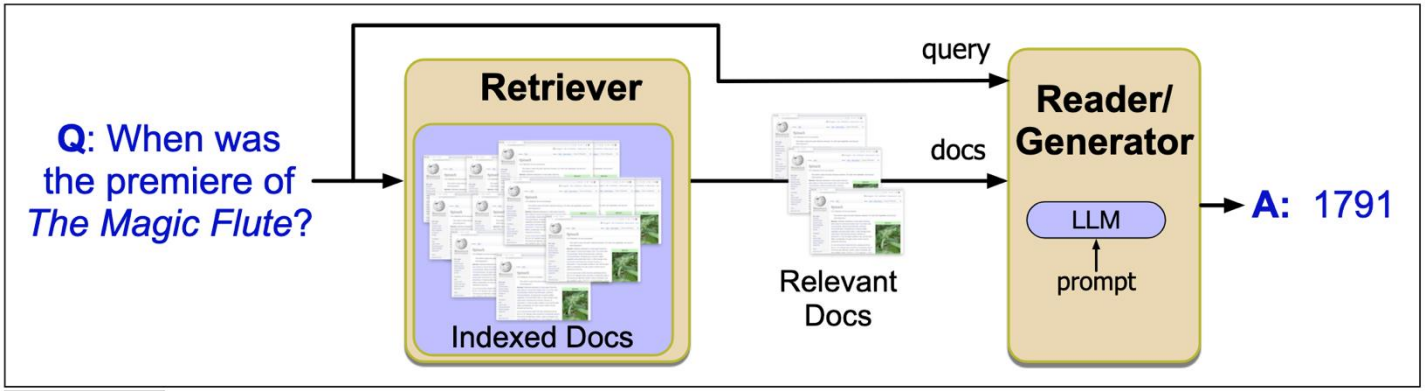
How can we incorporate:

- Domain-specific factual knowledge,
- In a way that a causal LM directly references,
- To improve task-specific performance,
- And reduce hallucinations?

Information retrieval provided in-context to **augment generation.**

How: Retrieval Augmented Generation (RAG)

Automatically search a database for relevant information and provide to the model in-context.



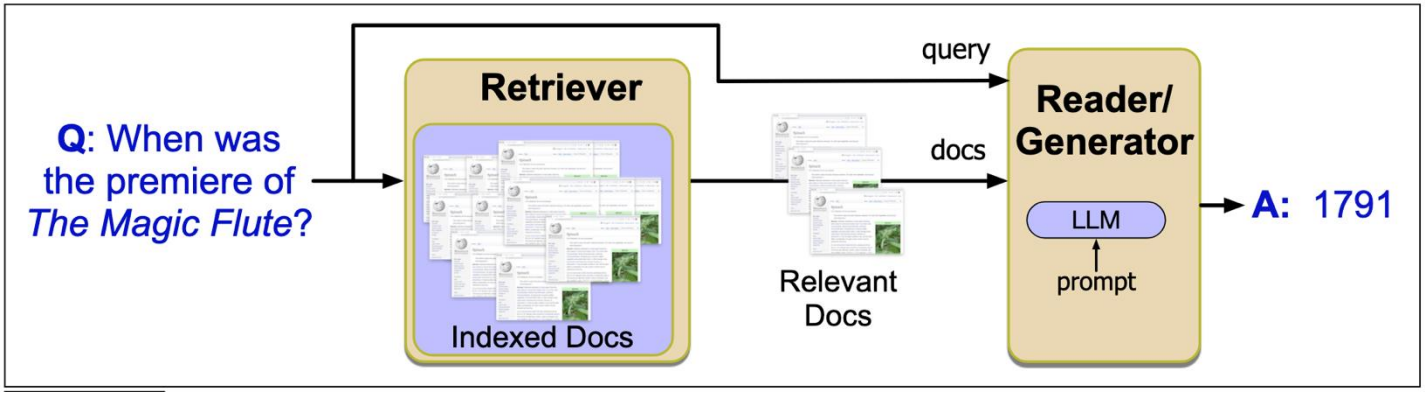
Schematic of a RAG Prompt

```
retrieved passage 1

retrieved passage 2

...

retrieved passage n

Based on these texts, answer this question:  Q: Who wrote
the book ‘‘The Origin of Species"?  A:
```

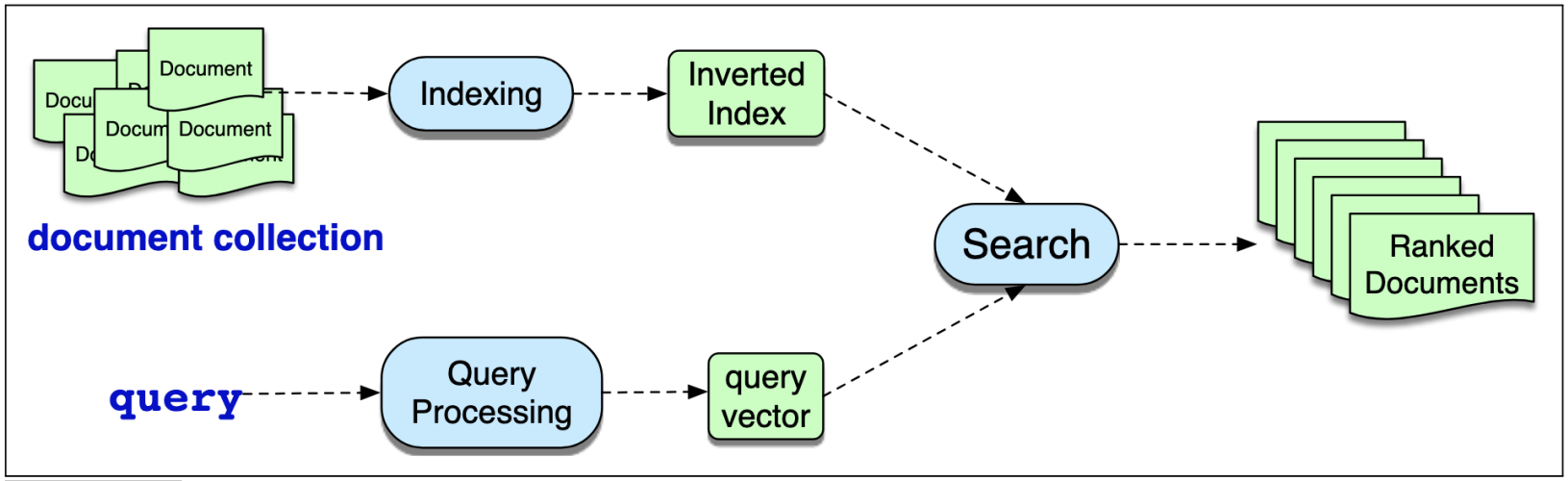
Prepending documents allows query to be contextualized with attention to documents.

Key Advantages of RAG

1. **Reduce hallucination** by providing curated and material directly in-context.
2. Can provide appropriate and relevant **citations** to improve **trust and verifiability**.
3. **Dynamic** information: Can incorporate new information post model-training cutoff.



Information Retrieval



- Traditional approaches look for matches on keywords using metrics like tf-idf (term frequency – inverse document frequency).
- But these approaches do not handle synonyms, word order, or contextual understanding.

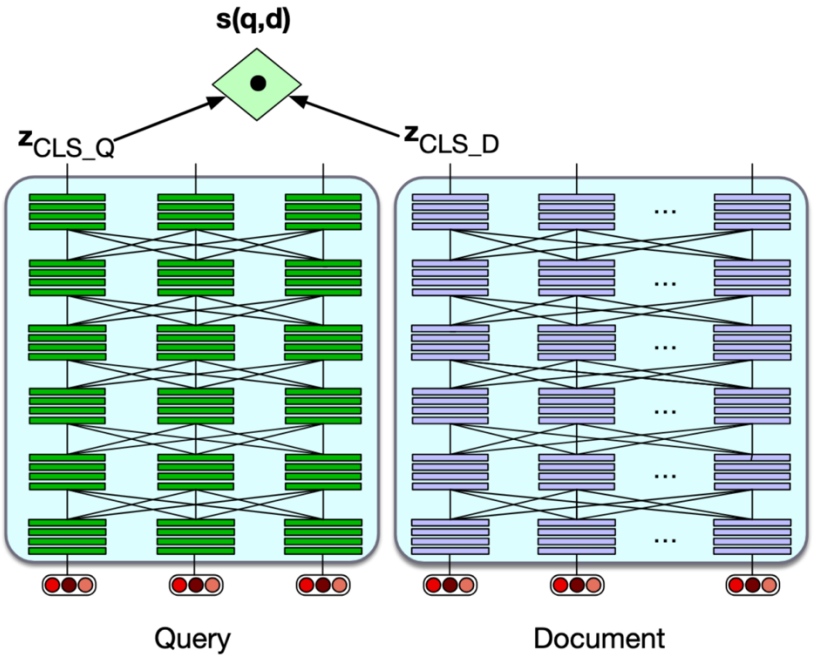
RAG Implementation with Dense Bi-Encoder Embeddings

In practice, the most common approach is to compute **dense embeddings** (numerical vectors) of both the query (prompt) and document (candidate for retrieval) using a pretrained encoder Transformer like BERT.

$$\mathbf{z}_q = \mathrm{BERT}_Q(\mathrm{q})[\mathrm{CLS}]$$

$$\mathbf{z}_d = \mathrm{BERT}_D(\mathrm{d})[\mathrm{CLS}]$$

$$\mathrm{score}(q,d) = \mathbf{z}_q \cdot \mathbf{z}_d$$



Select the highest scoring documents to add to the the RAG prompt.

Complexity of RAG with Dense Bi-Encoder Approach

$\boldsymbol{z}_d$ can be precomputed for all documents d in your database ahead of time.

At inference time with a query (prompt) q:

1. Compute $\boldsymbol{z}_q$ (forward propagation in encoder)
2. Compute $\mathrm{score}(q, d)$ for all documents d
3. Select highest scoring documents and pass to RAG prompt

Only done **once** before generation

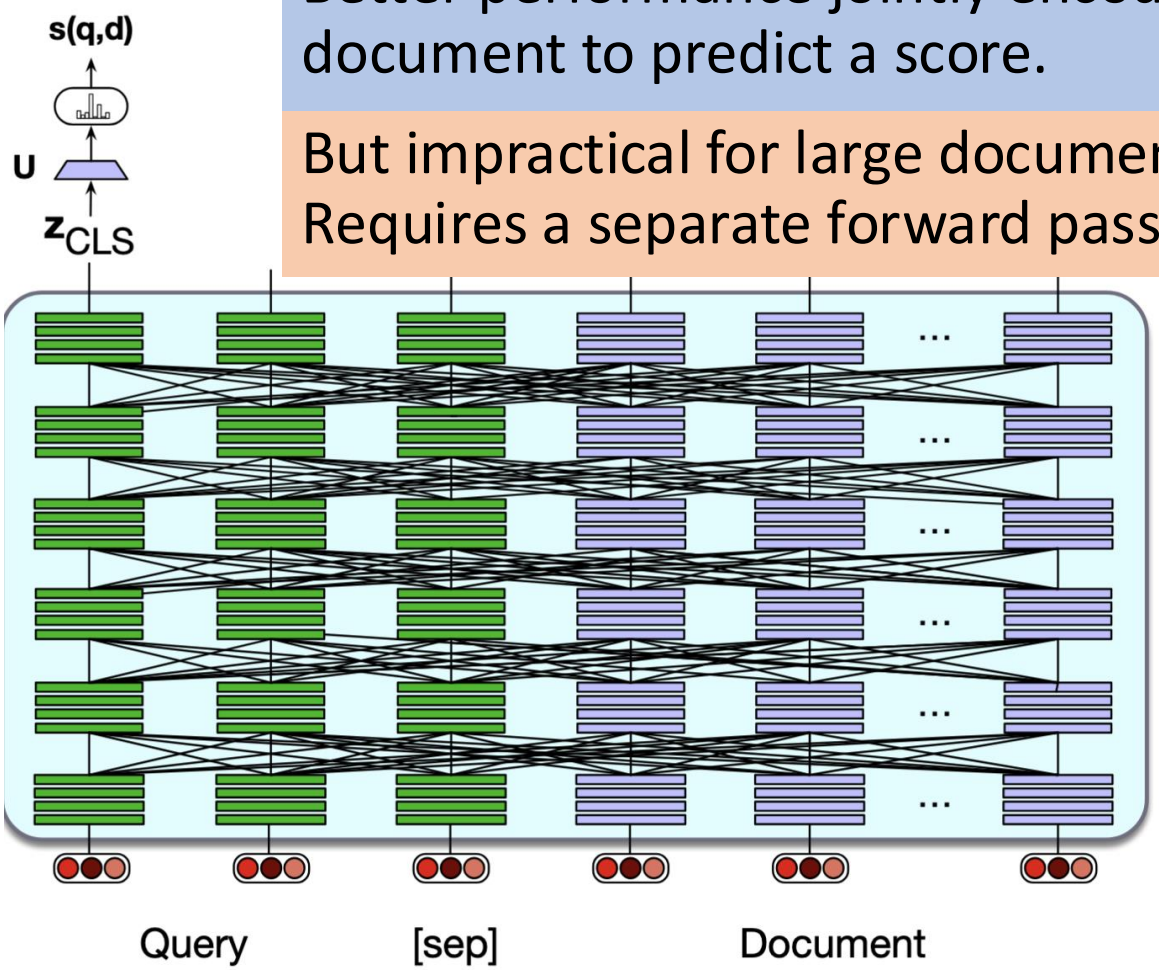
Can compute all scores in parallel

Linear scaling with number of documents in database

Less Common Alternative: Unified Encoder

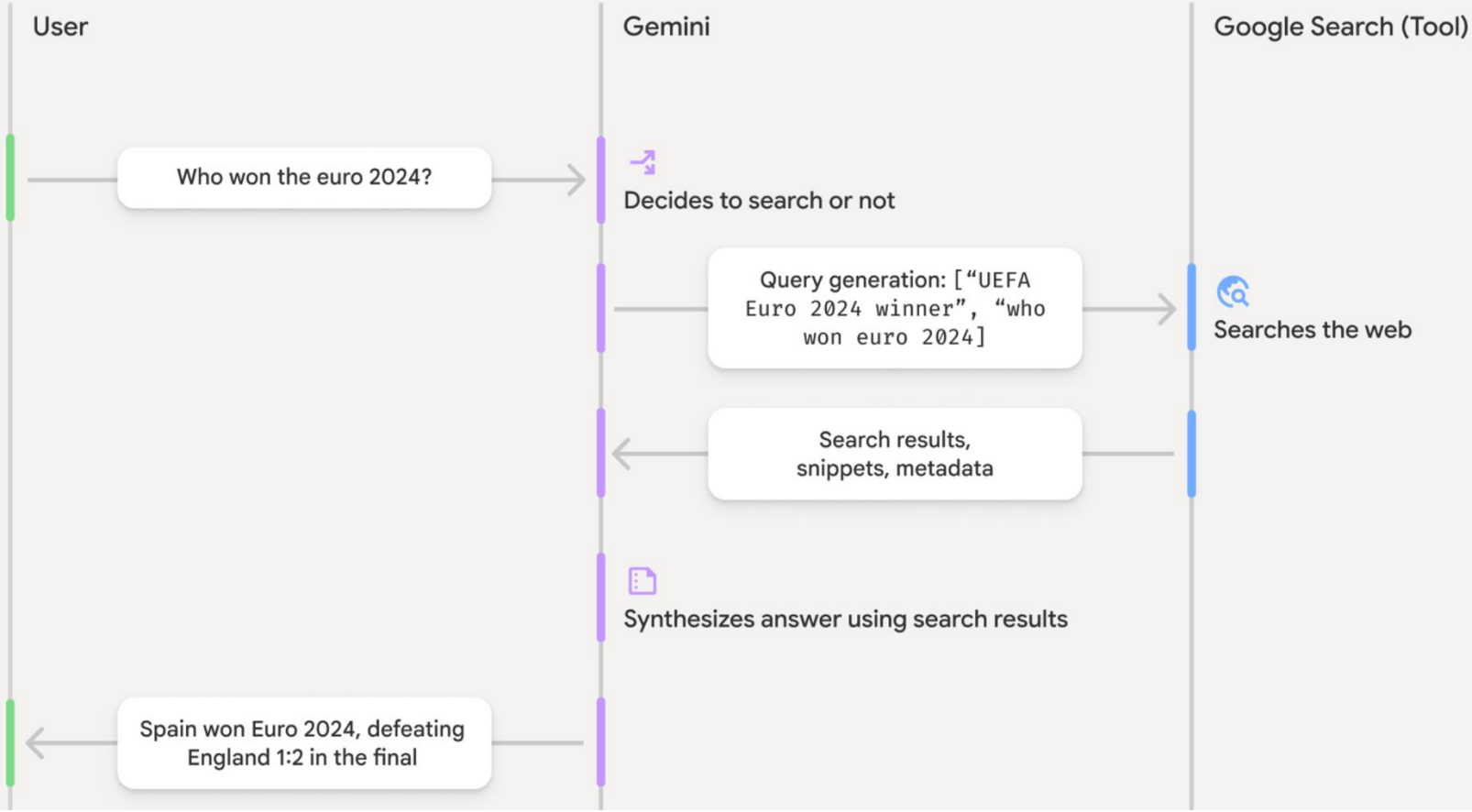
Better performance jointly encoding query and document to predict a score.

But impractical for large document collections. Requires a separate forward pass for each.



Web-Grounded Generation

Increasingly common to use existing internet search engines as a document retrieval tool to augment (or “ground”) generation.



ai.google.dev/gemini-api/docs/google-search (retrieved 9/30/2025)

“AI-Overview” in Search as Web-Grounded Generation Systems

how do google search ai summaries work?

AI Overview

Google's AI summaries, or AI Overviews, work **by using large language models (LLMs) to synthesize information from multiple webpages and other sources to generate a concise, conversational, and easy-to-understand answer at the top of the search results page**. The AI identifies key information, processes it, and then presents it along with citations to the original sources, adapting its responses based on user interactions and feedback to continuously improve. Content that is high-ranking, reliable, and addresses user intent is more likely to be featured in these AI-generated summaries.

how has google ai overview changed internet traffic

AI Overview

Google's AI Overview has **significantly decreased organic traffic and click-through rates (CTRs)** for many websites, especially those that rely on informational content. By